

	$0 = \Delta Q + (-200), \Delta Q = 200 \text{ J i.e. Net heat absorbed by gas} = 200 \text{ J.}$
17.	Ans: C $pV = nRT$ Hence work done by the gas $= p\Delta V = nR\Delta T$ $n = \frac{W}{R\Delta T} = \frac{20}{8.31(373 - 273)} = 0.02407 \text{ mol}$ Mass of helium gas present $= 0.02407 \times 4.0 = 0.096 \text{ g}$
18.	Ans: D